

HybridChain-OSINT OS Deployment Guide

This guide covers deploying HybridChain-OSINT OS as a system service for production use.

System Requirements

- **OS:** Linux (Ubuntu 20.04+, Debian 11+, CentOS 8+, or compatible)
- **Python:** 3.9 or higher
- **Memory:** Minimum 2GB RAM (4GB+ recommended for multi-user deployments)
- **Storage:** Minimum 10GB free space (scales with evidence storage)
- **Network:** Port 3000 (default, configurable)

Installation Methods

Method 1: Systemd Service (Recommended for Production)

1. Create System User

```
sudo useradd -r -m -s /bin/bash -d /opt/hybridchain-osint-os hybridchain
```

2. Install Application

```
# Clone repository
sudo su - hybridchain
git clone <repository_url> /opt/hybridchain-osint-os
cd /opt/hybridchain-osint-os

# Create virtual environment
python3 -m venv venv
source venv/bin/activate

# Install dependencies
pip install --upgrade pip
pip install -e .
```

3. Create Data Directories

```
sudo mkdir -p /var/lib/hybridchain
sudo mkdir -p /var/log/hybridchain
sudo chown -R hybridchain:hybridchain /var/lib/hybridchain
sudo chown -R hybridchain:hybridchain /var/log/hybridchain
sudo chmod 700 /var/lib/hybridchain # Protect sensitive data
```

4. Configure Environment

```
# Create environment configuration
sudo nano /opt/hybridchain-osint-os/.env
```

Add:

```
HYBRIDCHAIN_DATA_DIR=/var/lib/hybridchain
HYBRIDCHAIN_LOG_DIR=/var/log/hybridchain
HYBRIDCHAIN_PORT=3000
HYBRIDCHAIN_HOST=0.0.0.0
HYBRIDCHAIN_JWT_SECRET=<generate_secure_random_string>
HYBRIDCHAIN_JWT_EXPIRY=86400
HYBRIDCHAIN_NTP_ENABLED=true
HYBRIDCHAIN_MAX_UPLOAD_MB=100
```

5. Initialize System

```
sudo su - hybridchain
cd /opt/hybridchain-osint-os
source venv/bin/activate

# Initialize admin user
hybridchain-cli init-admin --username admin --password <secure_password>
```

6. Install Systemd Service

```
sudo cp /opt/hybridchain-osint-os/deployment/hybridchain.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable hybridchain
sudo systemctl start hybridchain
```

7. Verify Installation

```
# Check service status
sudo systemctl status hybridchain

# View logs
sudo journalctl -u hybridchain -f

# Test API
curl http://localhost:3000/api/health
```

Method 2: Docker Deployment (Coming Soon)

```
# Build image
docker build -t hybridchain-osint-os:latest .

# Run container
docker run -d \
  -p 3000:3000 \
  -v /var/lib/hybridchain:/data \
  --name hybridchain \
  hybridchain-osint-os:latest
```

Method 3: Multi-Instance Deployment

For running multiple isolated OSINT operations:

```
# Instance 1: Operation Alpha
HYBRIDCHAIN_DATA_DIR=/var/lib/hybridchain/alpha \
HYBRIDCHAIN_PORT=3001 \
hybridchain-server &

# Instance 2: Operation Beta
HYBRIDCHAIN_DATA_DIR=/var/lib/hybridchain/beta \
HYBRIDCHAIN_PORT=3002 \
hybridchain-server &
```

Or with systemd:

```
# Copy and modify service file for each instance
sudo cp /etc/systemd/system/hybridchain.service /etc/systemd/system/hybridchain-
alpha.service
sudo nano /etc/systemd/system/hybridchain-alpha.service
# Update ports, data directories, etc.

sudo systemctl enable hybridchain-alpha
sudo systemctl start hybridchain-alpha
```

Reverse Proxy Setup (Nginx)

For production deployments with SSL/TLS:

```

server {
    listen 443 ssl http2;
    server_name osint.example.com;

    ssl_certificate /etc/letsencrypt/live/osint.example.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/osint.example.com/privkey.pem;

    # Security headers
    add_header Strict-Transport-Security "max-age=31536000; includeSubDomains" always;
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;

    # Increase upload size for evidence files
    client_max_body_size 100M;

    location / {
        proxy_pass http://127.0.0.1:3000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}

# Redirect HTTP to HTTPS
server {
    listen 80;
    server_name osint.example.com;
    return 301 https://$server_name$request_uri;
}

```

Firewall Configuration

```

# UFW (Ubuntu/Debian)
sudo ufw allow 3000/tcp
sudo ufw enable

# Firewallld (CentOS/RHEL)
sudo firewall-cmd --permanent --add-port=3000/tcp
sudo firewall-cmd --reload

```

Backup Strategy

Automated Backup Script

Create `/opt/hybridchain-osint-os/scripts/backup.sh` :

```
#!/bin/bash
BACKUP_DIR="/backup/hybridchain"
DATA_DIR="/var/lib/hybridchain"
DATE=$(date +%Y%m%d_%H%M%S)

mkdir -p "$BACKUP_DIR"

# Backup blockchain data
tar -czf "$BACKUP_DIR/hybridchain_$DATE.tar.gz" \
  "$DATA_DIR/chain.jsonl" \
  "$DATA_DIR/public_chain.jsonl" \
  "$DATA_DIR/users.json" \
  "$DATA_DIR/community_members.json" \
  "$DATA_DIR/keys/"

# Backup evidence files separately (can be large)
tar -czf "$BACKUP_DIR/evidence_$DATE.tar.gz" \
  "$DATA_DIR/evidence/"

# Keep last 30 days of backups
find "$BACKUP_DIR" -name "*.tar.gz" -mtime +30 -delete

echo "Backup completed: $DATE"
```

Add to cron:

```
sudo crontab -e
# Daily backup at 2 AM
0 2 * * * /opt/hybridchain-osint-os/scripts/backup.sh
```

Monitoring

Health Check Endpoint

```
curl http://localhost:3000/api/health
```

Response:

```
{
  "status": "healthy",
  "chain_height": 1234,
  "public_chain_height": 890,
  "user_count": 15,
  "community_members": 45,
  "ntp_status": "synchronized",
  "storage_gb_used": 2.5
}
```

Prometheus Integration (Optional)

Add metrics endpoint by installing:

```
pip install prometheus-flask-exporter
```

Configure in `osint_chain/api/app.py` :

```
from prometheus_flask_exporter import PrometheusMetrics
metrics = PrometheusMetrics(app)
```

Troubleshooting

Service Won't Start

```
# Check logs
sudo journalctl -u hybridchain -n 100

# Verify permissions
ls -la /var/lib/hybridchain
ls -la /opt/hybridchain-osint-os

# Test manual start
sudo su - hybridchain
cd /opt/hybridchain-osint-os
source venv/bin/activate
hybridchain-server
```

High Memory Usage

```
# Monitor process
top -p $(pgrep -f hybridchain-server)

# Limit memory (systemd)
sudo nano /etc/systemd/system/hybridchain.service
# Add under [Service]:
MemoryLimit=2G
```

Database Corruption

```
# Verify chain integrity
hybridchain-cli verify

# Restore from backup
sudo systemctl stop hybridchain
cd /var/lib/hybridchain
sudo cp chain.jsonl chain.jsonl.corrupted
sudo tar -xzf /backup/hybridchain/hybridchain_YYYYMMDD_HHMMSS.tar.gz
sudo systemctl start hybridchain
```

Security Hardening

File Permissions

```
# Restrict access to data directory
sudo chmod 700 /var/lib/hybridchain

# Protect private keys
sudo chmod 600 /var/lib/hybridchain/keys/*.pem

# Protect configuration
sudo chmod 600 /opt/hybridchain-osint-os/.env
```

Network Security

- Use TLS/SSL (reverse proxy)
- Implement rate limiting
- Enable fail2ban for authentication attempts
- Restrict API access by IP (if possible)

Application Security

- Rotate JWT secrets periodically
- Enforce strong password policies
- Regular security updates: `pip install --upgrade -e .`
- Monitor access logs for suspicious activity

Scaling

Horizontal Scaling (Load Balancing)

For high-traffic deployments:

1. **Read Replicas:** Run multiple read-only instances
2. **Write Master:** Single instance for evidence submission
3. **Load Balancer:** Nginx/HAProxy for distribution

Vertical Scaling

- Increase RAM for larger chains
- SSD storage for faster access
- Multi-core CPU for concurrent API requests

Maintenance

Regular Tasks

- **Daily:** Check logs, verify backups
- **Weekly:** Review chain integrity (`hybridchain-cli verify`)
- **Monthly:** Update dependencies, security patches
- **Quarterly:** Review user access, rotate JWT secrets

Updating

```
sudo systemctl stop hybridchain
sudo su - hybridchain
cd /opt/hybridchain-osint-os
git pull
source venv/bin/activate
pip install --upgrade -e .
exit
sudo systemctl start hybridchain
```

Support

For deployment issues:

- Check [docs/TROUBLESHOOTING.md](#) (../docs/TROUBLESHOOTING.md)
- Review [CONTRIBUTING.md](#) (../CONTRIBUTING.md)
- Open an issue on GitHub

Remember: HybridChain-OSINT OS handles sensitive investigation data. Always follow security best practices and comply with your organization's data protection policies.